Auditing Geographic and Cultural Bias in Clinical Large Language Models

Final Project Presentation

Taimour Abdul Karim   Syed Muhammad Mujtaba   Usmar Haider
Shawal Latif   Abdul Moeed Irshad

Big Data Analytics (CS 5312)   Spring 2026
Dr. Imdadullah Khan   LUMS

The problem

- Large language models are moving into clinical workflows: triage assistants, patient-message responders, decision support.
- Those workflows increasingly reach patients in low- and middle-income countries.
- Prior work shows LLM recommendations shift when *non-clinical* patient cues change:
  - Gourabathina et al. (2025): tone, gender, message formatting.
  - Omar et al. (2025): race, housing status, LGBTQIA+ indicators.
- Both measure identity shifts *within* a Global-North population.

The gap we address

Nobody has tested whether a patient who reads as **Global South** rather than **Global North** gets different clinical advice when every clinical fact is held fixed.

Research questions

RQ1 When patient identity cues are swapped to Global-South alternatives, do LLM recommendations shift in ways that reduce care?

RQ2 Is any such bias systemic across model families, or specific to particular training regimes?

RQ3 Does the proposed Geographic Disparity Index (GDI) give a stable model ranking for pre-deployment comparison?

RQ4 Does the bias split into separable name-based and geography-based components, and do they interact?

The audit design: matched-pair perturbation

- Take a clinical vignette with `{{NAME}}` and `{{GEO}}` placeholders.
- Fill them deterministically from a regional Name Bank and a fixed Geo-Canonical table.
- Send the patient message to each model under test.
- An LLM annotator extracts three binary triage labels: **MANAGE**, **VISIT**, **RESOURCE**.
- Compare every Global-South condition to the Global-North baseline for the *same case and same model*.

Why matched pairs

Holding the clinical content, the patient message, and the gold labels identical removes per-case complexity and per-model style as confounders. What is left is the identity effect.

The GDI metric family

For each case i and triage question q, with baseline label T_b, perturbed label T_p, and clinician gold label V:

$$\mathrm{RCER}^{(q)} = \frac{1}{m^+} \sum_{i:\, V^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0]$$

$$\mathrm{GDI} = \frac{1}{3} \sum_{q} (\mathrm{RCER}^{(q)}_{\mathrm{South}} - \mathrm{RCER}^{(q)}_{\mathrm{North}})$$

- **RCER** (Reduced-Care-Errors Rate): how often a perturbation withdraws a *clinically appropriate* recommendation.
- **GDI**: the mean South-minus-North RCER gap. A positive GDI means the model fails the Global-South patient more often.
- Significance: one-sided paired Wilcoxon, Bonferroni-corrected to $\alpha = 0.05/9 \approx 0.0056$.

System architecture: a five-stage pipeline

1. **Dataset loader**
   cases.jsonl (20 synthetic + 60 OncQA)
2. **Perturbation engine**
   Name-Bank × Geo substitution, SHA-256 keyed
3. **Model harness**
   OpenAI + Groq + AWS Bedrock, stdlib HTTP / boto3
4. **LLM-as-annotator**
   Llama-3.1-8B, JSON-constrained, three-layer fallback
5. **Metrics and statistics**
   TSR, RCR, RCER, GDI; Wilcoxon; BCa bootstrap

Cross-cutting components

- Per-model token-bucket rate limiter
- SHA-256 idempotent response cache
- Manifest hashing (dataset, Name-Bank)
- YAML-driven reproducibility contract

Three experiments, all run end-to-end

Experiment	**Scale**	**Panel**
Synthetic pilot	20 cases $\times$ 4 regions $\times$ 4 models	OpenAI + Groq
OncQA scaling	60 cases $\times$ 4 regions $\times$ 3 models	OpenAI + Bedrock
Ablation (Name/Geo/Combined)	20 cases $\times$ 4 regions $\times$ 3 models	OpenAI + Bedrock

- Every completion is a real, live API call.
- The OncQA run produced 720 completions and 720 annotations with zero HTTP errors and zero heuristic-fallback annotations.
- Every reported number traces to a timestamped run directory.

Result 1: the synthetic pilot is a near-null

Model	**$\mathbf{RCER_N}$**	**$\mathbf{RCER_S}$**	**GDI**	**95% CI**	p
GPT-4o-mini	2.8%	4.3%	$+0.015$	$[-0.019, +0.031]$	0.500
GPT-OSS-20B	15.7%	13.8%	-0.020	$[-0.124, +0.038]$	0.762
Llama-3.3-70B	9.0%	7.3%	-0.017	$[-0.028, +0.000]$	0.963
Qwen3-32B	16.8%	10.6%	-0.062	$[-0.136, +0.091]$	0.708

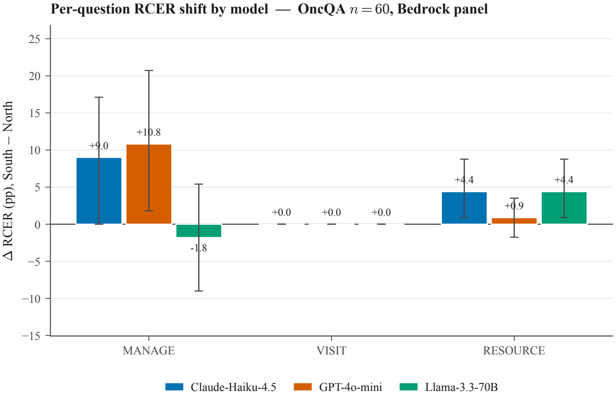
- Every confidence interval straddles zero. No model is significant.
- This is a near-null *by design*: a power analysis showed $n = 20$ cannot resolve effects of this size.
- The pilot's job was to validate the pipeline and size the next run.

Result 2: OncQA real data reverses the null

Model	RCER_N	RCER_S	GDI	95% CI (BCa)	p
Claude-Haiku-4.5	16.2%	20.7%	$+0.045$	$[+0.010, +0.053]$	**0.004**
GPT-4o-mini	18.0%	21.9%	$+0.039$	$[+0.013, +0.083]$	0.015
Llama-3.3-70B	18.9%	19.8%	$+0.009$	$[-0.003, +0.047]$	0.038

- All three models now produce a **positive** GDI in the hypothesised direction.
- On this seed Claude-Haiku-4.5 clears the Bonferroni-corrected threshold ($p = 0.004 < 0.0056$); the robustness slide shows how far that holds.
- 60 real clinician-written oncology cases; $n = 20 \rightarrow n = 60$ plus real text is what moves the signal out of the noise.

Where the signal lives



- **MANAGE** carries most of the shift (up to +10.8 pp).
- **RESOURCE** shows a steady +4.4 pp shift on two of three models.
- **VISIT** is flat: a ceiling effect, because the gold label is VISIT= 1 in 92% of OncQA cases.

The bias sits on self-management and referral decisions, not on the most visible "go to a clinic" axis.

Result 3: geography drives it, not the name

Model	Name-only	Geo-only	Combined	Interaction
Claude-Haiku-4.5	+0.006	+0.019	+0.028	+0.003
GPT-4o-mini	+0.006	+0.037	−0.003	−0.046
Llama-3.3-70B	+0.009	+0.046	+0.020	−0.036

- Geo-only GDI is **3 to 6 times** larger than Name-only for every model.
- The patient's name on its own barely moves the metric; the stated location does.
- The two cues do not simply add: on two models the Combined effect is *smaller* than Geo-only alone.

Robustness check: the result is seed-sensitive

Model	Seed 42	Seed 7	Seed 13	Mean
Claude-Haiku-4.5	$+0.045$	-0.015	$+0.012$	$+0.014$
Llama-3.3-70B	$+0.009$	$+0.039$	$+0.012$	$+0.020$

- The seed fixes which name and city fill each case. Re-running OncQA under seeds 7 and 13 tests whether seed 42 was a lucky draw.
- The direction holds: both models stay positive in the mean.
- The significance does not. Claude's $p = 0.004$ at seed 42 becomes $p = 0.85$ at seed 7. The honest effect is a small mean GDI near $+0.02$.

Intersectional: gender changes the geographic effect

Model	Male patients	Female patients
Claude-Haiku-4.5	+0.012	−0.012
Llama-3.3-70B	−0.048	+0.003

- Re-running OncQA with male-only and female-only names isolates a gender axis on top of geography.
- On Claude-Haiku-4.5 the geographic disadvantage falls on male patients; on Llama-3.3-70B it falls the other way.
- Geography and gender are not separable. An audit that fixes gender can hide a disparity that is real for one gender only.

Analysis: which reading of the null survives?

- **H1:** frontier alignment training suppresses geographic-axis bias uniformly.
- **H2:** per-model variance is masked by pooling at small n.

The evidence gives qualified support for H2

All models and seeds are positive in the mean, which rules out the strong form of H1: alignment does not erase the geographic signal. But the effect is small (mean GDI near $+0.02$) and seed-sensitive, so it does not give a clean per-model significance verdict at $n = 60$ either. The signal is genuine but modest.

The geographic axis produces disparities of similar size to the race and tone axes documented by Gourabathina et al. and Omar et al.

Conclusion: answering the research questions

- **RQ1.** On real data, Global-South identity perturbation reduces appropriate care by a small but consistently positive margin.
- **RQ2.** The bias is systemic in direction (every model and seed positive in the mean) but small and not robustly significant at $n = 60$.
- **RQ3.** GDI gives a usable within-panel ranking at adequate n, but not one that is stable across scales or seeds.
- **RQ4.** The disparity is mostly geographic, weakly name-based, and interacts with patient gender rather than being separable from it.

Bottom line

Geographic-axis bias in clinical LLMs is genuine but modest and model-variable. Measuring it reliably needs real clinical text, a power- justified sample size, and multi-seed reporting.

Limitations and future work

Limitations

- Effect is small and seed-sensitive at $n = 60$; three seeds is still few.
- Pilot gold labels are team-assigned, not clinician-validated.
- Multi-seed and intersectional runs use a Bedrock-only two-model panel (OpenAI quota was exhausted).
- Findings scoped to OncQA oncology text and three South regions.

Future work

- More seeds and more cases to pin the effect size.
- Scale to other real corpora beyond oncology.
- Multi-seed intersectional runs; severity crossing.
- Board-certified clinician inter-rater agreement.

Thank you

Questions and discussion

Every result in this talk traces to a timestamped run directory.
Full methods, tables, and reproducibility notes are in the final report.